

Relativity and Electromagnetism 2026 spring semester

Meeting times

Wed 08:30-09:45 S9 1.1

Wed 10:00-11:15 S9 1.1

Thu 08:30-09:45 S9 1.1

Thu 14:30-15:45 S9 3.1 (~~Feb 12~~, Feb 26, Mar 12, Mar 26)

Schedule

Week 01 (Feb 11, 12): WW~~T~~~~T~~

Week 02 (Feb 18, 19): WW~~T~~–

Week 03 (Feb 25, 26): WWTT

Week 04 (Mar 04, 05): WW~~T~~–

Week 05 (Mar 11, 12): WWTT

Week 06 (Mar 18, 19): WW~~T~~–

Week 07 (Mar 25, 26): WW~~T~~~~T~~

Week 08 (Apr 01, 02): WW~~T~~–

Week 09 (Apr 22, 23): WW~~T~~–

Week 10 (Apr 29, 30): WW~~T~~–

Week 11 (May 06, 07): WW~~T~~–

Week 12 (May 13, ~~14~~): WW––

Note: W indicates a Wednesday class and T indicates a Thursday class. – indicates no class.

Textbook

David J. Griffiths, “*Introduction to Electrodynamics*” (fifth or fourth edition), Cambridge University Press

Course plan

Week 1: Overview, electrostatics, magnetostatics (§ 2.1, 2.2, 5.1, 5.2, 5.3)

Week 2: Electrodynamics: the full set of Maxwell’s equations (§ 7.1, 7.2, 7.3)

Week 3: Potentials, work and energy (§ 2.3, 2.4, 5.4)

Week 4: Conservation laws (§ 8.1, 8.2.2, 9.2)

Week 5: Gauge transformations, potential formulation, retarded potentials (§ 10.1, 10.2)

Week 6: Relativity: Lorentz transformations (§ 12.1)

Week 7: Relativistic mechanics (§ 12.2)

Week 8: Relativistic electrodynamics (§ 12.3)

=== 2 week easter break ===

Week 9: Conductors, separation of variables, multipole expansion (§ 2.5, 3.2, 3.3, 3.4)

Week 10: Dielectrics, magnetic materials (§ 3.2, 4.1, 4.2, 4.3, 4.4, 6.1, 6.2, 6.3, 6.4)

Week 11: Radiation (§ 11.1, 11.2)

Week 12: Revision, discussion of exam format

Note: This plan is tentative. Section numbers from the textbook are indicated in parentheses.

Office and contact

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Preferred means of contact: over Ufora

Teaching assistants

Robin Chan
Matthias Vereecken
Robbe Brants

Exercises

Wed Feb 11 10:00: Ex 1
Wed Feb 18 10:00: Ex 2
Thu Feb 26 14:30: Ex 3
Thu Mar 05 08:30: Ex 4
Thu Mar 12 14:30: Ex 5
Thu Mar 19 08:30: Ex 6
Thu Mar 26 14:30: Ex 7
Thu Apr 02 08:30: Ex 8
=== 2 week easter break ===
Thu Apr 30 08:30: Ex 9
Thu May 07 08:30: Ex 10

Note: Typically the last class of every week is an exercise session. There are some exceptions. Additional exercises are covered during the class.

Homework assignments

HW 1 due Thu Mar 26 (tentative)
HW 2 due Thu Apr 23 (tentative)
HW 3 due Thu May 07 (tentative)
HW 4 due Thu May 21 (tentative)

Grading

- 40% homework assignments (4 × 10% each)
- 60% final exam on Wed Jun 17 (theory and exercises)

Note: You will need to pass the final exam (independently of your homeworks) to pass the course!